

DATA SCIENCE & AI LAB · MAY 2026 · GROUP 1

Multimodal Damage Assessment for Insurance Claims

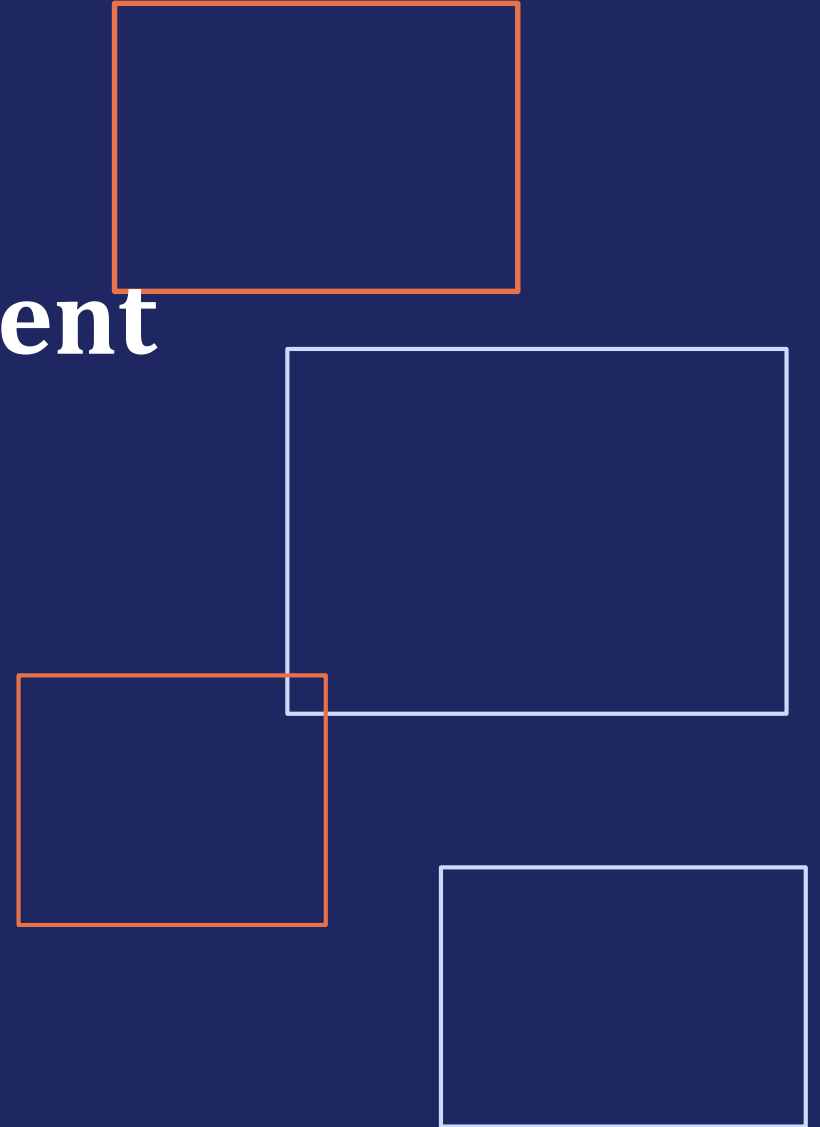
MILESTONE 2: DATASET PREPARATION

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From Milestone 1 to a Training-Ready Dataset

Milestone 1 defined a multi-agent RAG system that reads damage photos and policy PDFs, detects and classifies damage, retrieves relevant policy clauses, and drafts a preliminary claim report with a LangGraph orchestrator routing each claim across four specialist agents and escalating low-confidence cases to human review.

MILESTONE 2 OBJECTIVES

- 1 Dataset verification & download**
Identify provenance, confirm licensing, download every dataset each agent needs.
- 2 Vision data preparation**
Clean, remap, dedupe, augment and split VehiDE into training-ready artefacts.
- 3 Policy corpus construction**
Author, chunk, embed and index a synthetic policy corpus for the Policy Agent.

FOUR-AGENT ORCHESTRATION

Damage Agent

YOLO detection of 6 damage classes

Severity Agent

Bounding-box area-ratio scoring

Policy Agent

RAG over synthetic policy corpus (MCP tool)

Report Agent

LLM-based preliminary claim report

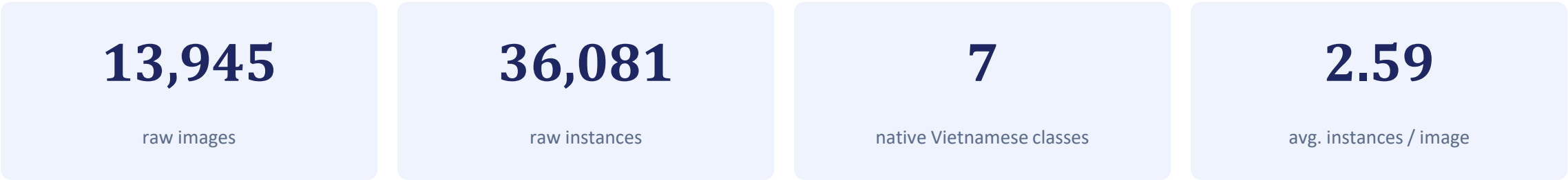
Low-confidence outputs escalate to a human review queue.

Six Datasets, Four Agents

Dataset	Feeds	Role	Status
VehiDE (primary)	Damage + Severity	6-class YOLO fine-tuning & severity calibration	Ready
CarDD	Damage	Pixel-level segmentation masks	Pending access
Car Damage Severity	Severity	Human-labelled Minor/Moderate/Severe ground truth	Ready
COCO Car Damage	Damage (comparison)	Architecture sanity-check	Ready
Synthetic policy PDFs	Policy Agent	RAG retrieval corpus	Ready
Synthetic incident descriptions	Report + Eval	Ground-truth incident/report pairs	Ready

VehiDE is the largest publicly available annotated vehicle-damage dataset (13,945 images, 36,081 raw instances), selected as the primary training source. CarDD and the Severity dataset will be integrated once the training on model performs poorly.

VehiDE: Structure & Scale



ANNOTATION SCHEMA (VIA-format JSON)

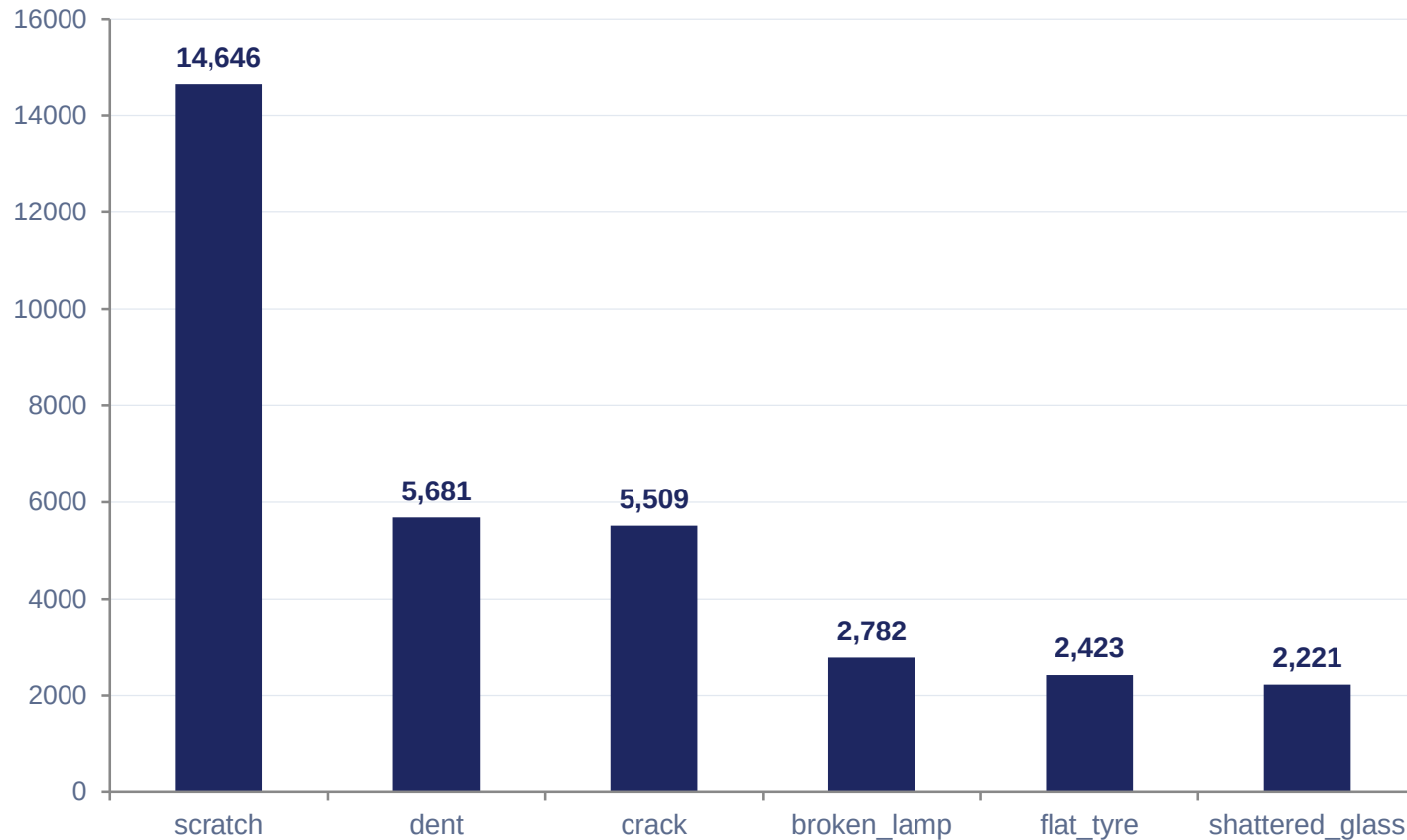
```
{
  "<file>.jpg": {
    "regions": [
      {"all_x": [...], "all_y": [...],
       "class": "rach"}, ...
    ]
  }
}
```

Bounding boxes are derived from each polygon's [x, y, width, height] extent.

VIETNAMESE → PROJECT CLASS MAPPING

Native	Project class
tray_son (paint_scratches)	scratch
mop_lom (dents)	dent
rach (torn_body)	crack
be_den (broken_lamp)	broken_lamp
thung (puncture)	flat_tyre
vo_kinh (broken_glass)	shattered_glass
mat_bo_phan (lost_parts)	EXCLUDED

Class Distribution & Imbalance

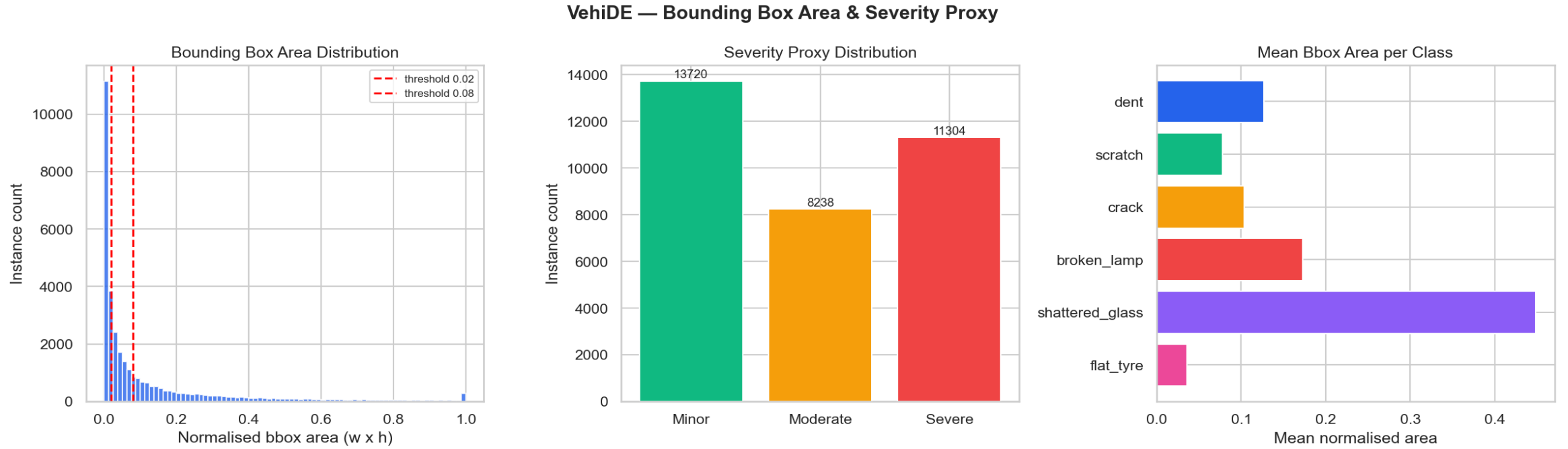


6.59 : 1

imbalance ratio · scratch vs. shattered_glass

- scratch dominates at 44.0% of retained instances
- shattered_glass is rarest at 6.7%
- torn_body is termed as crack class
- Mitigated via class-weighted loss & oversampling of minority classes

Bounding-Box Area & Severity Proxy



Minor

13,720 41.3%

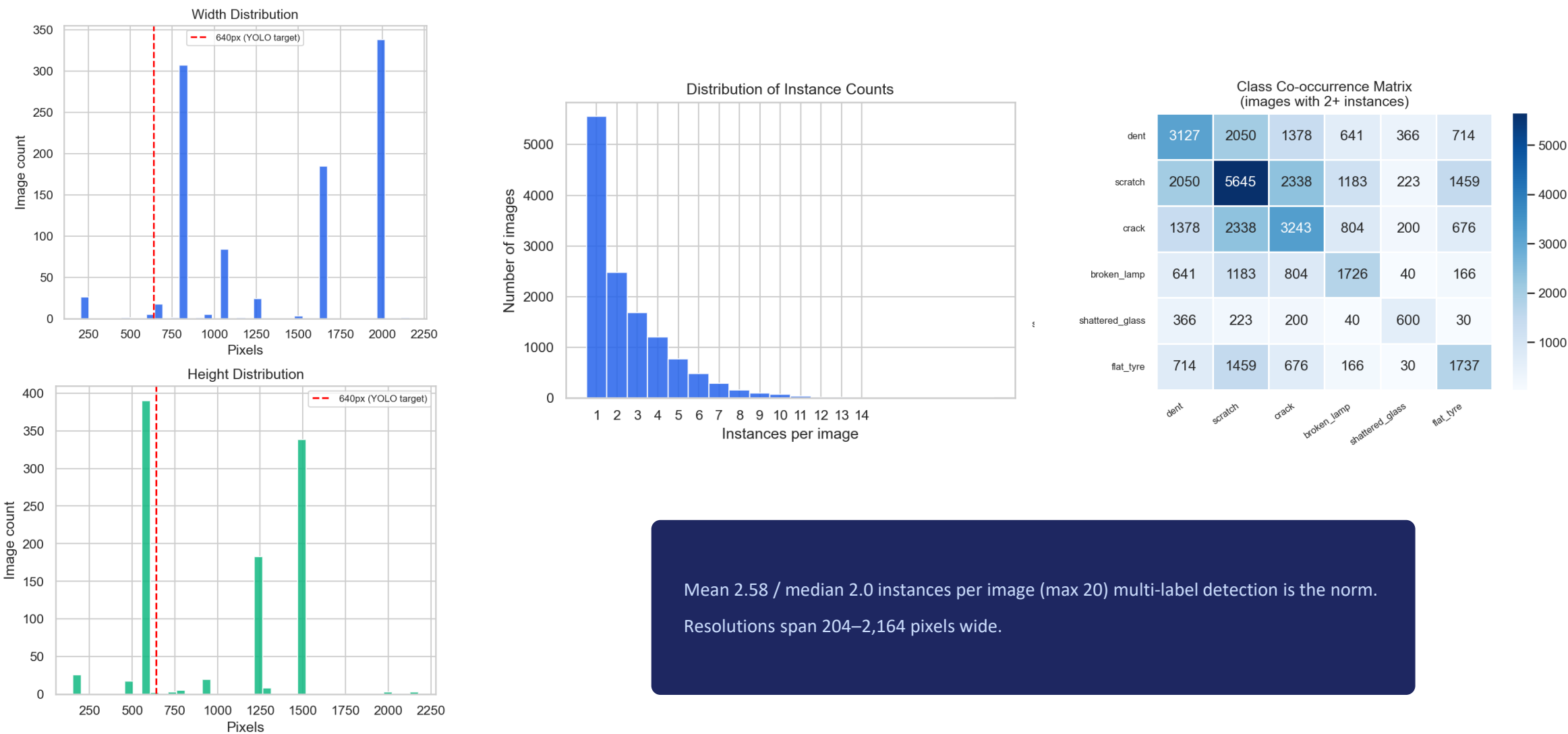
Moderate

8,238 24.8%

Severe

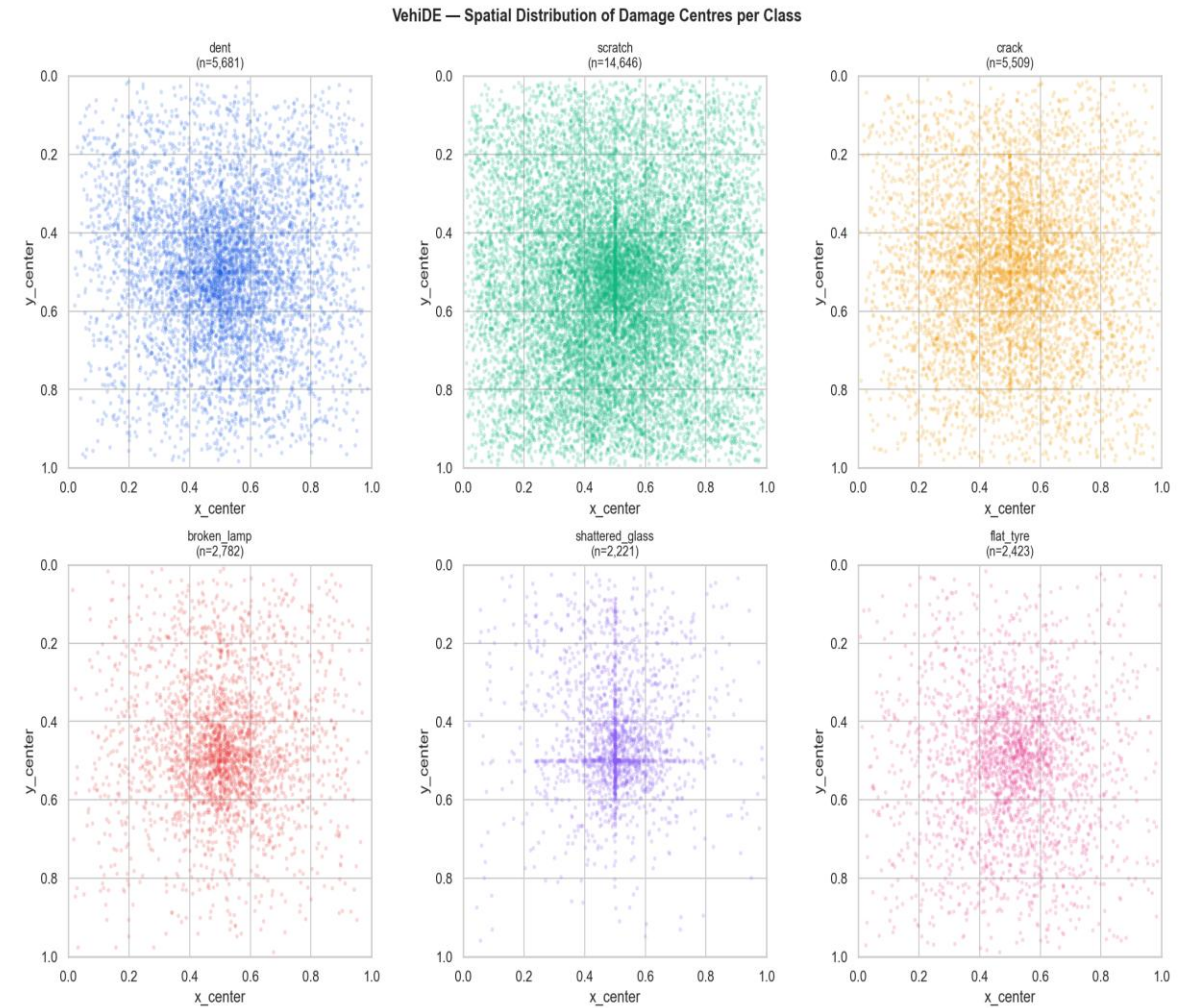
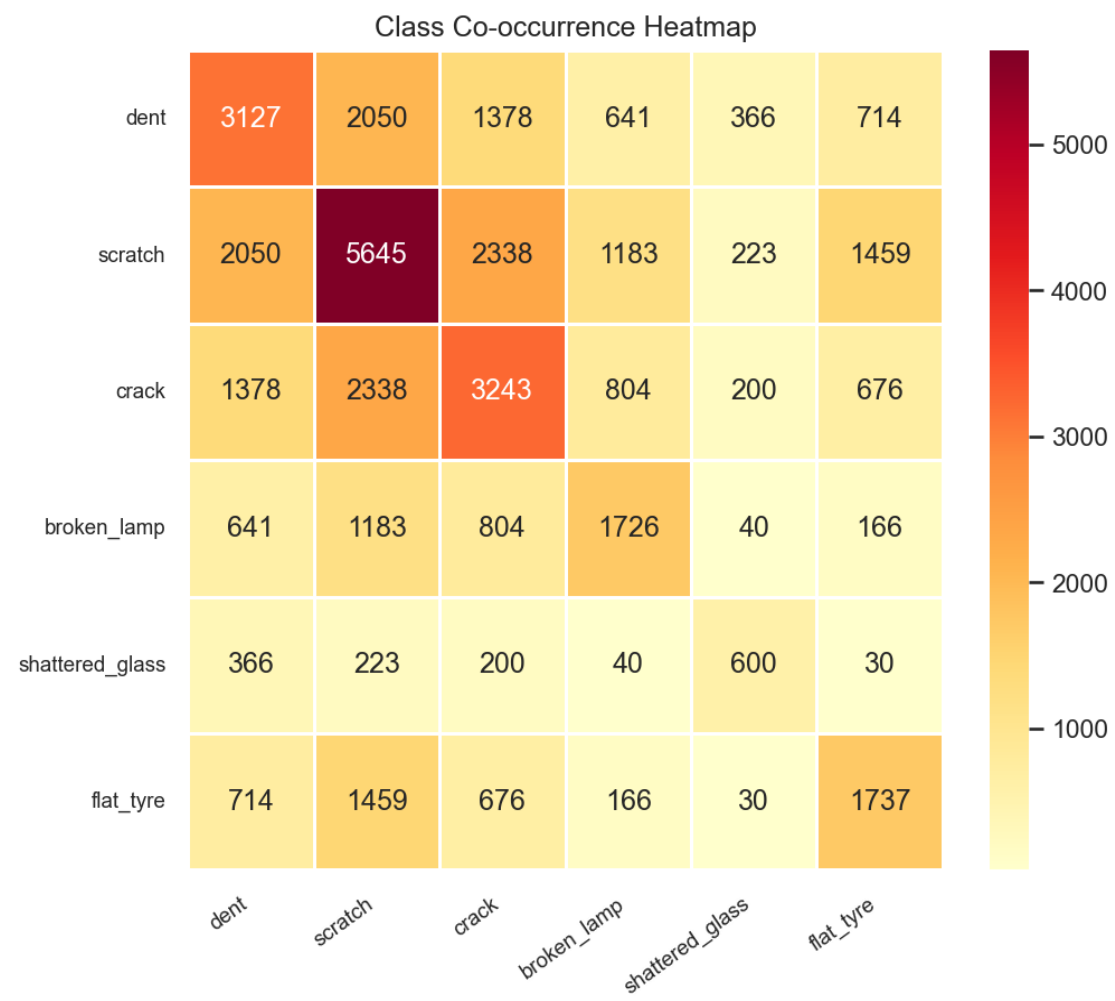
11,304 34.0%

Multi-Instance Images & Resolution Spread



Mean 2.58 / median 2.0 instances per image (max 20) multi-label detection is the norm.
Resolutions span 204–2,164 pixels wide.

Where and With What Damage Appears



Licensing, Privacy, Ethics & Reproducibility

Source & Licensing

VehiDE (Apache-2.0), CarDD & Kaggle sets under academic/community terms.
IRDAI policy PDFs used only as structural reference.

Privacy

Face / license-plate detector run across all 13,655 images, 0 flagged. Detector stays in the pipeline for future additions.

Data Quality

0 corrupt files, 0 orphan images/annotations, 1 malformed region corrected, 18 exact & 272 near-duplicates removed.

Ethics & Bias

VehiDE is Southeast-Asia-sourced, Indian vehicle types & conditions may be underrepresented. Mitigated via augmentation & flagged as a domain-shift risk.

From Raw Download to Training-Ready Images

1

Corrupt-file check

0 corrupt images found

2

Class remap (7→6)

torn_body → crack
lost_parts excluded

3

YOLO format verify

Polygon → [x,y,w,h] bbox, normalised

4

Letterbox resize

640×640, grey pad (114,114,114)

5

Annotation spot-check

5% sample rendered & visually verified

6

PII detection

Face/plate detector 0 flagged

7

Deduplication

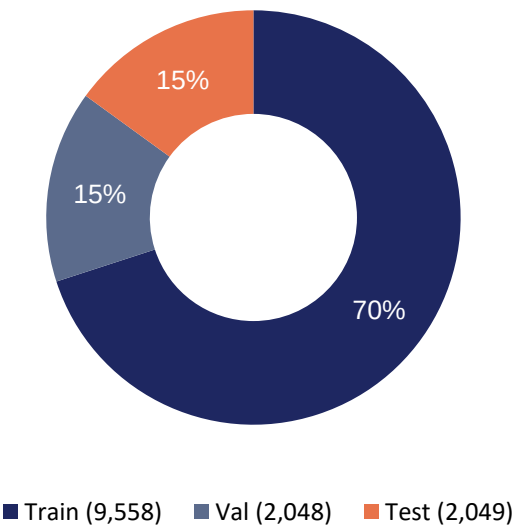
18 exact & 272 near-duplicates removed

8

Stratified split

70 / 15 / 15 → train / val / test

Balanced, Leakage-Checked, Training-Ready



13,655 images · 32,672 instances
70 / 15 / 15 stratified split

AUGMENTATION (TRAIN-TIME ONLY)

- Horizontal flip (p=0.5)
- Mosaic (4-image combine)
- HSV brightness / saturation jitter
- Small rotation & translation
- Motion blur mobile camera shake
- JPEG re-compression (quality 75)

CLASS-TARGETED WEIGHTS (cls_pw)

Class	Weight
scratch	1.0
dent	2.6
crack	2.7
broken_lamp	5.3
flat_tyre	6.0
shattered_glass	6.6

RAG Corpus for the Policy Agent

5

synthetic PDFs

179

indexed chunks

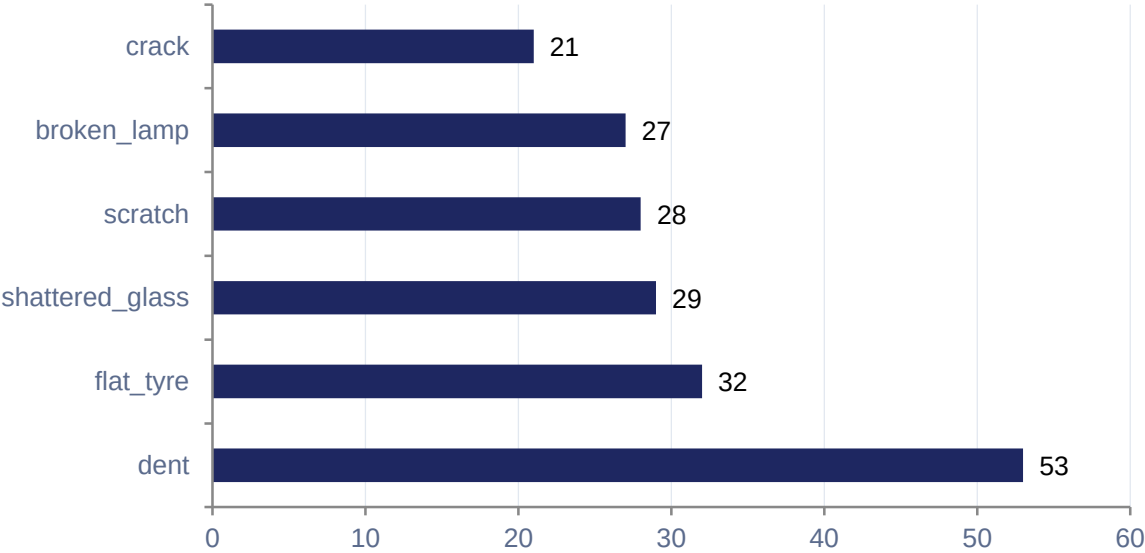
300 / 40

chunk size / overlap (tokens)

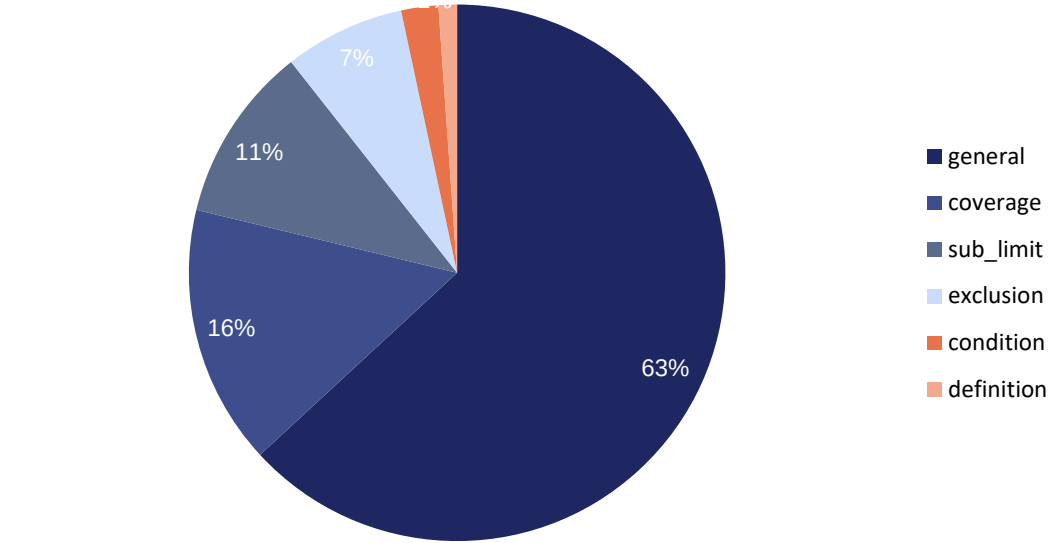
6

damage classes tagged

CHUNKS PER DAMAGE CLASS



CLAUSE TYPE DISTRIBUTION



Ready for Model Training

13,655

images

32,672

retained instances

6

damage classes

179

policy chunks indexed

The data is ready for training immediately:

- Train / val / test splits finalised, leakage-checked, and logged as file lists.
- ChromaDB policy index (collection policy_clauses, 179 chunks) built and queryable.
- CarDD segmentation augmentation & severity-proxy calibration remain open, pending data access.

Open Items Going Into Milestone 3

Challenge	Status / Resolution
Class imbalance	Class-weighted loss & oversampling will be applied while training
CarDD dataset access	Manual licensing form required
Geographic bias in VehiDE	SE-Asia-sourced; mitigated via augmentation, flagged as domain-shift risk
Annotation format mismatch	VehiDE (YOLO bbox) vs. CarDD (COCO polygon) conversion script drafted, will be used if CarDD dataset is used.

What's Ready, What's Next

DELIVERABLES PRODUCED

- VehiDE EDA notebook + preprocessing script
- Cleaned & split image / label sets (train/val/test)
- Class remap lookup & YOLO config (damage.yaml, nc=6)
- Synthetic policy corpus + ChromaDB vector index (179 chunks)
- COCO Car Damage EDA (complete, used for comparison)
- Milestone 2 report

PLANNED FOR MILESTONE 3

- Select model architecture: YOLO11m-seg vs. YOLOv8m-seg baseline
- Define LangGraph orchestrator state schema & MCP tool contracts
- Run YOLO baseline (50 epochs) establish mAP@50 & per-class F1
- Wire up Policy Agent's FastMCP retrieval tool; run precision check
- Validate Report Agent prompt template on 5 sample incident/image pairs
- And if the model performs poorly, then integrate CarDD data into the training dataset.

Thank You

Questions & Discussion

Github: [HiveCase/Group-1-DS-and-AI-Lab-Project](#)
Course: DS & AI Lab · May 2026 · Group 1

